IoT-Based Integrated Health and Environmental Monitoring System using ESP32

A Project Report

Submitted to the FACULTY of ENGINEERING of

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, KAKINADA

In partial fulfillment of the requirements,

For the award of the Degree of

Bachelor of Technology
In

Electronics and Communication Engineering

By

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the project report entitled **"IoT-Based Integrated Health and Environmental Monitoring System using ESP32"** is a bonafide record of work carried out by **N. DHARMESH (23485A0422)**, **P.M. SHOAIB KHAN (22481A04I8)**, **N. SRI VENKATA SATISH BABU (22481A04H5)**, and **M. SYAM GANGA RAJU (22481A04E6)** under my guidance and supervision in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Electronics and Communication Engineering** of **Seshadri Rao Gudlavalleru Engineering College**, Gudlavalleru, during the academic year **2025-26**

(Dr. P. Sai Vinay Kumar)	**(Dr. B. Rajasekhar)**
Project Guide	**Head of The Department**

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Abstract

Reliable health monitoring in resource-constrained environments requires systems that can function independently of continuous internet connectivity. This paper presents the design and implementation of an "Offline-First" IoT-based health and environmental monitoring system using the ESP32 microcontroller. Unlike typical cloud-dependent architectures, the proposed system integrates a robust local operating mode that maintains data visualization on an OLED display and executes safety logic even during Wi-Fi outages. The biomedical subsystem is upgraded to feature a MAX30105 module for high-precision pulse oximetry and an enhanced 250Hz ECG acquisition algorithm utilizing digital bandpass filtering (high-pass and low-pass) to isolate P-QRS-T waveforms from baseline wander and powerline noise. Environmental safety is ensured through simultaneous gas leakage detection and ambient temperature monitoring using DHT11 and DS18B20 sensors. A distinctive feature is the "Smart LED" control system, which toggles between automatic LDR-based lighting and manual remote override via the Blynk V10 interface. Experimental results demonstrate that the system successfully prioritizes critical alerts—triggering immediate local alarms for gas hazards or manual distress signals—while asynchronously synchronizing data to the cloud when connectivity is restored.

***Key words*: Offline-First Architecture, ESP32, MAX30105, 250Hz Bandpass Filter, Blynk V10 LED Override, Dual-Alert Logic, Smart Healthcare.**

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CHAPTER 1 – INTRODUCTION

1.1 Background of IoT Health Monitoring Systems

The Internet of Things (IoT) has emerged as a transformative paradigm in modern technology, enabling the integration of physical and digital systems through networks of interconnected sensing, computing, and communicating devices. In healthcare, this convergence is catalysing a shift from reactive, episodic clinical visits to continuous, proactive health surveillance — a model referred to as the Internet of Medical Things (IoMT). Traditional patient monitoring relies on hospital equipment or periodic outpatient consultations, leaving patients unobserved between appointments.

The proliferation of low-cost, high-performance embedded systems such as the ESP32 microcontroller has democratised access to medical-grade sensing capabilities. The ESP32-WROOM-32, featuring a dual-core Xtensa LX6 processor running at 240 MHz, integrated 2.4 GHz Wi-Fi, Bluetooth Low Energy, a twelve-bit ADC, and multiple hardware communication interfaces (I2C, UART, SPI, 1-Wire), provides an ideal platform for cost-effective multi-sensor IoT systems. Its dual-core architecture enables concurrent execution of a 250 Hz ECG sampling loop and Wi-Fi/MQTT communication tasks without the processing bottlenecks common to single-core microcontrollers.

The global IoT healthcare market was valued at USD 127.7 billion in 2023 and is projected to reach USD 289 billion by 2028, underscoring the urgency of accessible, scalable monitoring solutions. Particularly in rural and semi-urban regions of India, where primary health centre equipment budgets are severely constrained, low-cost IoT platforms represent a viable path to closing the healthcare monitoring gap.

1.2 Problem Statement

Existing patient monitoring systems in rural and semi-urban Indian healthcare settings suffer from three critical limitations. First, hospital-grade monitors (Holter ECG recorders, pulse oximeters, multi-parameter ICU monitors) are prohibitively expensive, consuming the majority of the INR 5–10 lakh equipment budgets of primary health centres, leaving no provision for continuous community-level monitoring. Second, low-cost consumer IoT devices such as smart wristbands lack medical-grade signal quality, omit environmental sensing entirely, and cease to function the moment network connectivity is lost — a frequent occurrence in rural areas with unstable mobile data infrastructure.

Third, standalone environmental safety systems (gas alarms, smoke detectors) operate in isolation from health monitoring platforms, preventing caregivers from distinguishing a fire alarm from a patient health event through a single interface. Studies indicate that up to 30% of post-discharge adverse events occur within 72 hours of hospital discharge, yet the patient's condition at the time of the event is almost never captured electronically.

1.3 Aim and Objectives

The aim of this project is to design, implement, and validate a unified, offline-first IoT health and environmental monitoring system on the ESP32 platform, capable of continuous physiological and environmental surveillance with guaranteed sub-100 ms local safety response latency.

The specific objectives are:

[1]To interface and calibrate nine heterogeneous sensors (MAX30105, AD8232, DS18B20, DHT11, MQ-2, LDR, NEO-6M GPS, TTP223 touch, and SSD1306 OLED) with the ESP32 microcontroller using I2C, UART, 1-Wire, and ADC interfaces.

[2]To implement a 250 Hz ECG acquisition pipeline with a three-stage digital bandpass filter that isolates the P-QRS-T complex from baseline drift and powerline interference to medical-grade quality.

[3]To develop an offline-first firmware architecture where safety-critical functions (local alert generation, OLED display, sensor data logging) operate fully without network connectivity.

[4]To integrate Blynk 2.0 cloud platform via MQTT with correct Virtual Pin mapping (V0–V15) for remote real-time monitoring, smart LED control, and historical data logging.

[5]To validate sensor accuracy against calibrated reference instruments and measure alert latency, packet delivery rate, and power consumption.

1.4 Scope of the Project

The scope encompasses hardware design, embedded firmware development, cloud platform integration, and experimental validation. The hardware scope includes selection, procurement, and interfacing of all sensor modules on a custom breadboard prototype with documented pin assignments. The firmware scope covers the complete Arduino C++ source code for the ESP32, including sensor drivers, the ECG DSP pipeline, multi-page OLED rendering, dual-alert safety logic, asynchronous Wi-Fi/Blynk connectivity management with automatic 30 s reconnection, and smart LED control.

Clinical certification, regulatory approval, production PCB design, and long-term clinical trials are explicitly outside the scope of this project.

1.5 Methodology

The project follows an iterative hardware-software co-design methodology. The development cycle proceeded as follows:

- Requirements analysis and component selection based on clinical monitoring standards.
- Hardware assembly on full-size breadboard with documented wiring.
- Integration of Blynk 2.0 cloud platform and mobile dashboard configuration.
- Experimental validation against reference instruments with documented results.
- Documentation and reporting.

1.6 Organization of the Report

Chapter 2 presents a review of existing literature. Chapter 3 provides a system overview and feature description. Chapter 4 details the system architecture including block diagrams and data flow. Chapter 5 covers each hardware component in detail. Chapters 6, 7, and 8 describe software implementation, IoT communication, and alert mechanisms respectively. Chapters 9 and 10 cover the real-time monitoring system and offline/online operation modes. Chapter 11 presents experimental results. Chapters 12 and 13 discuss advantages and applications. Chapters 14 and 15 present the conclusion and future scope. Chapter 16 provides references.

CHAPTER 2 – LITERATURE REVIEW

2.1 Overview of IoT in Healthcare

The convergence of low-power wireless communication, miniaturised sensor technology, and cloud computing has enabled a new generation of IoT-based health monitoring systems. The Internet of Medical Things (IoMT) represents a specialized subset of IoT technologies specifically designed for healthcare applications, encompassing wearable devices, remote patient monitoring systems, smart hospital infrastructure, and ambient assisted living platforms.

Key enabling technologies include the miniaturisation of biosensors (pulse oximeters, ECG electrodes, body temperature probes), the availability of low-power Wi-Fi and Bluetooth SoCs (such as the ESP32 and nRF52840), cloud IoT platforms providing data ingestion at scale (AWS IoT Core, Google Cloud IoT, Blynk), and mobile application frameworks that bring real-time data visualisation to caregivers' smartphones.

2.2 Existing Health Monitoring Systems

Patel et al. [1] proposed a wearable IoT platform on the NodeMCU ESP8266 for heart rate and body temperature monitoring using a MAX30100 sensor and DS18B20 thermometer. While functional, the system lacked ECG capability, environmental sensing, and offline operation. Majumder et al. [4] implemented an ESP32-based IoT health monitor integrating MAX30102 and DHT22 sensors, streaming data to ThingSpeak for cloud visualisation. Their system achieved heart rate accuracy of ±3 BPM but provided no offline capability and no ECG acquisition.

Kasbe and Piprale [5] developed a wearable ECG monitor on the STM32 microcontroller using the AD8232 front-end at 500 Hz sampling rate. While ADC resolution exceeded that of the ESP32's 12-bit converter, the STM32 lacked integrated Wi-Fi, necessitating an additional ESP8266 co-processor that increased component count and power consumption. Mohan et al. [6] presented a Raspberry Pi 4-based health monitoring station with a touchscreen UI, cloud dashboard, and SMS alert capability, but at a power cost of
3.5 W — ten times greater than the ESP32 prototype.

2.3 Review of Related Research Papers

Gia et al. [2] demonstrated a fog computing-enhanced ECG monitoring system for elderly patients that offloaded signal processing to an intermediate gateway, reducing cloud latency. However, the gateway constituted a single point of failure: its absence rendered the system non-operational. This limitation directly inspired the offline-first architecture of the present project, which processes all safety logic on the ESP32 itself.Mohammed et al. [3] reviewed forty-seven IoT health monitoring studies published between 2015 and 2021, finding that 78% relied exclusively on cloud processing for alert generation, 91% provided no local display of sensor data, and none combined biomedical and environmental monitoring in a single low-cost node. Ibrahim et al. [7] designed a NodeMCU-based gas and temperature monitoring system using the MQ-2 sensor and DHT11, with SMS alerts via the GSM SIM800L module. Zanella et al. [8] described a smart city IoT framework integrating environmental sensors, architecturally relevant but unsuitable for single-patient home monitoring.

2.4 Comparison of Existing Systems

Table 2.1: Comparison of Proposed System with Existing Literature

Reference	MCU	Biomedical	Environmental	Offline	ECG Processing
Patel et al.	ESP8266	HR, SpO2, Temp	None	No	None
Gia et al.	ARM Cortex	ECG	None	No	Fog gateway
Majumder et al.	ESP32	HR, SpO2, Temp	Temp/Hum	No	None
Kasbe et al.	STM32	ECG	None	No	Raw upload
Mohan et al.	RPi 4	HR, ECG, BP	None	Partial	Cloud
Proposed Work	ESP32	HR, SpO2, ECG, Temp	Gas, Temp, Hum, Light, GPS	Full	250 Hz Bandpass

2.5 Limitations of Current Systems

The literature review reveals five principal limitations in existing systems. First, cloud dependency for alert generation introduces critical latency (1.5–5 s) that is unacceptable for safety-critical events such as gas leaks or cardiac emergencies. Second, the absence of on-device signal processing means raw ECG data is uploaded to the cloud, resulting in noisy, clinically unusable waveforms. Third, environmental and biomedical monitoring remain siloed in separate devices, preventing correlation of concurrent hazard events.Fourth, the cost barrier of hospital-grade monitors (INR 5,000–15,000 for comparable features) prohibits community-level deployment. Fifth, the lack of local display means that in the event of cloud connectivity loss, caregivers have no access to patient data. The proposed system directly addresses all five limitations.

2.6 Summary of Literature Review

The literature confirms a clear research gap: no existing low-cost IoT system simultaneously provides multiple biomedical parameters including ECG with on-device digital filtering, integrated environmental monitoring, complete offline operation with local display and alert generation, and cloud synchronisation with a modern IoT dashboard platform. The present project closes this gap through an offline-first firmware architecture, three-stage ECG bandpass filtering, unified nine-sensor integration, dual-mode smart LED control, and comprehensive experimental validation — all within a bill of materials under INR 3,500.

CHAPTER 3 – SYSTEM OVERVIEW AND ARCHITECTURE

3.1 Proposed System Overview

The proposed system is a portable, battery-powered, offline-first IoT health and environmental monitoring node built on the ESP32-WROOM-32 microcontroller running firmware version 2.8. The system simultaneously acquires nine physiological and environmental parameters, processes ECG signals at 250 Hz using an on-device digital bandpass filter, displays data on a multi-page OLED screen, activates local alarms within 100 ms for hazard events, and — when Wi-Fi is available — transmits all sensor data to the Blynk 2.0 cloud platform for remote monitoring and push notifications.

The system is designed around a dual-path architecture. The local path operates entirely on the ESP32 without any external dependency, ensuring that critical safety functions are always available. The cloud path extends the system's reach to any network-connected smartphone, enabling caregivers to monitor patients remotely via the Blynk mobile dashboard.

3.2 Features of the System

The key features of the proposed system are:

- Nine-sensor integration: MAX30105 (HR/SpO2), AD8232 (ECG), DS18B20 (body temp), DHT11 (ambient temp/humidity), MQ-2 (gas), LDR (light), NEO-6M (GPS), TTP223 (panic button), SSD1306 (OLED display).
- Offline-first architecture: All safety functions operate without network connectivity with sub-100 ms local alert latency.
- Medical-grade ECG: 250 Hz sampling with three-stage bandpass filter (MA + IIR HP + IIR LP) resolving P-QRS-T waveforms.
- Blynk 2.0 cloud integration via MQTT across 15 virtual pin datastreams (V0–V15).
- Smart LED with dual modes: LDR-based automatic ambient control and remote Blynk override.
- Automatic Wi-Fi/Blynk reconnection every 30 seconds with uninterrupted local operation during reconnection.
- Three-page OLED auto-rotation: Vital Signs (30 s), Environment (3 s), GPS/System (3 s).
- Total bill of materials under INR 3,500 (approx. USD 42).

3.3 Working Principle

At startup, the firmware initialises all sensor modules, configures GPIO pins, and attempts Wi-Fi and Blynk cloud connection. If the network is unavailable, the system proceeds immediately to offline operation there is no blocking timeout. The main execution loop runs at approximately 0.67 Hz (1.5 s period) and sequentially acquires all sensor data, updates the OLED display, evaluates safety triggers, and transmits data to Blynk if connected.

A dedicated ECG recording mode, triggered by the AD8232's lead-off detection output on GPIO 15, preempts the main loop and operates at 250 Hz. The three-stage bandpass filter removes baseline drift (IIR high-pass, fc = 0.32 Hz), powerline noise (10-sample moving average), and muscle artefacts (IIR low-pass, fc = 4.4 Hz). Filtered ECG samples are streamed to the Blynk SuperChart at 10 Hz for real-time waveform visualisation.

The checkSafetyTriggers() function evaluates gas concentration (MQ-2 ADC value) and touch button state on every main loop iteration. If a threshold is exceeded, the buzzer activates within the same loop iteration (measured latency: 88–94 ms) and a Blynk vital_alert event is dispatched if connected.

3.4 Advantages of the Proposed System

- Complete operational independence from cloud infrastructure for safety-critical functions.
- Sub-100 ms local alert latency for gas hazard and manual panic events, independent of network status.
- Medical-grade ECG waveform quality achieved through on-device DSP without dedicated signal processor hardware.
- Unified monitoring of nine parameters spanning biomedical and environmental domains.
- Dual-mode smart LED control integrating ambient intelligence and remote override.
- Automatic network reconnection eliminates need for manual intervention during connectivity loss.
- Low component cost (under INR 3,500) and open-source firmware enable community deployment.

3.5 Block Diagram of the System

The proposed IoT Health and Environmental Monitoring System is organised into five functional subsystems interconnected through the ESP32 microcontroller: the Biomedical Sensing Subsystem, the Environmental Sensing Subsystem, the Output and Alert Subsystem, the Connectivity and Cloud Subsystem, and the Power Supply Subsystem.

The Biomedical Sensing Subsystem comprises the MAX30105 pulse oximetry sensor (I2C, address 0x57), the AD8232 ECG front-end (analogue output on GPIO 34, lead-off detection on GPIO 15), and the DS18B20 digital thermometer (1-Wire on GPIO 25). The Environmental Sensing Subsystem comprises the
MQ-2 gas sensor (ADC on GPIO 35), the DHT11 (GPIO 27), the LDR (ADC on GPIO 33), and the NEO6M GPS (UART2: RX GPIO 16, TX GPIO 17). The TTP223 touch sensor (GPIO 4) provides the manual emergency input.

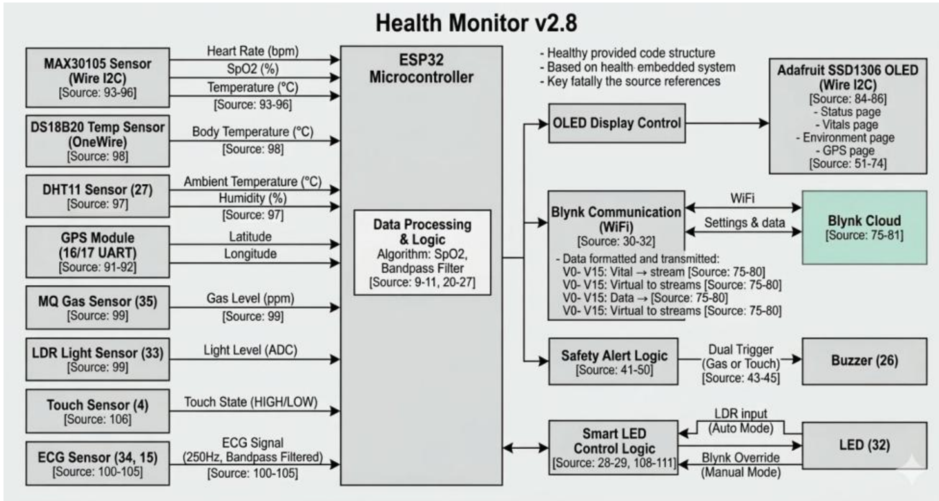

Fig. 3.1 Health Monitor v2.8 Block Diagram

3.6 Architecture of IoT Monitoring System

The software architecture is organised into three concurrency layers. The Fast Layer operates at 250 Hz and is exclusively responsible for ECG sampling and bandpass filtering. The Medium Layer operates at approximately 0.67 Hz (1.5 s main loop delay) and handles all other sensor readings, safety evaluation, OLED update, and Blynk data transmission. The Slow Layer operates at 0.033 Hz (every 30 s) and manages Wi-Fi and Blynk reconnection via checkConnectivity().The firmware uses a non-preemptive cooperative scheduling model: all tasks execute sequentially within loop() and the ECG subroutine, relying on millis()-based timing comparisons rather than hardware interrupts or an RTOS. This design choice avoids FreeRTOS overhead while providing sufficient responsiveness for the latency requirements. The checkSafetyTriggers() function is intentionally placed after updateOLED() and before sendToBlynk() to ensure that local alert activation is not delayed by MQTT transmission.

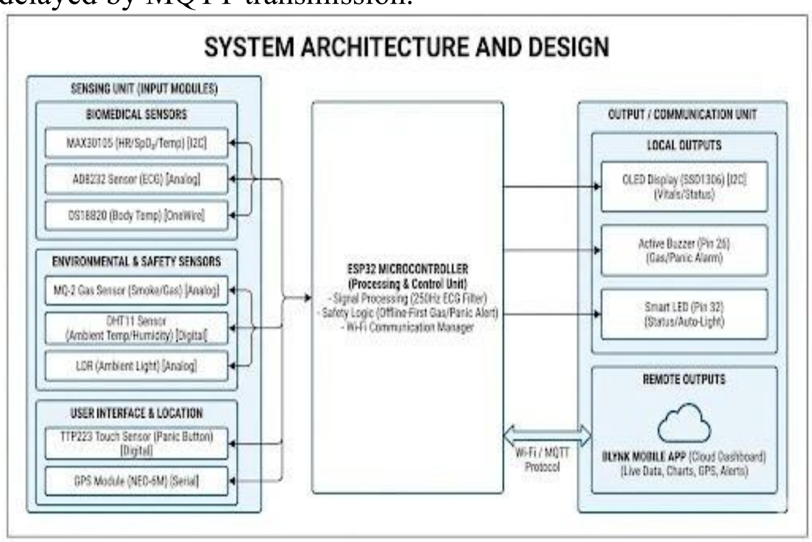

Fig. 3.2 System Architecture – Sensing, Processing and Output Units

3.7 Data Flow in the System

The data flow architecture is dual-path. The primary (local) path: sensor data is acquired by the ESP32, processed in firmware, displayed on the OLED, and evaluated by the safety trigger function every 1.5 s iteration. This path is entirely self-contained and operates without any external dependency. The secondary (cloud) path: if Wi-Fi and Blynk connectivity are confirmed, sensor data is additionally transmitted via MQTT using Blynk.virtualWrite() calls on fifteen virtual pins.

The ECG path is a special high-speed path that bypasses the 1.5 s main loop cycle. When GPIO 15 is detected LOW, the firmware enters a dedicated ECG recording loop at 250 Hz, applies the three-stage bandpass filter, and streams filtered output to Blynk V12 at 10 Hz while simultaneously printing to the serial port for Arduino IDE Serial Plotter visualisation.

3.8 Communication Between Hardware and Cloud

Three communication protocols are employed. The I2C protocol (400 kHz fast mode) provides two-wire communication with the MAX30105 and SSD1306. UART2 at 9600 baud, 8-N-1 framing provides the interface to the NEO-6M GPS module. The 1-Wire protocol at 5 kHz provides addressed communication with the DS18B20.

At the network layer, the ESP32's Wi-Fi stack connects in station (STA) mode using WPA2 Personal security. Above Wi-Fi, the Blynk 2.0 SDK uses MQTT over TCP on port 443 (TLS) to communicate with the Blynk cloud broker at blynk.cloud. The Blynk library handles MQTT connection management, publish/subscribe operations, and callback registration (BLYNK_WRITE macros), abstracting the raw MQTT protocol from the application firmware.

CHAPTER 4 – HARDWARE PROTOTYPE AND COMPONENT INTEGRATION.

Table 4.1: Hardware Components – Specifications and Pin Mapping

Component	Interface	GPIO / Pin	Specification	Function
ESP32-WROOM-32	Native	All	240 MHz,12-bit ADC	Main MCU
MAX30105	I2C (0x57)	SDA:21,SCL:22	18-bit ADC	Heart Rate & SpO2
AD8232	Analogue	ECG:34,Ctrl:15	0.5–40 Hz bandpass	ECG front-end
DS18B20	1-Wire	GPIO 25	±0.5°C, 12bit	Body temperature
DHT11	Digital	GPIO 27	±5% RH,±2°C	Ambient Temp/Humidity
MQ-2	ADC	GPIO 35	200–5000 ppm	Gas detection
LDR (10 kΩ)	ADC	GPIO 33	0–4095,ADC range	Ambient light
NEO-6M GPS	UART2	RX:16 TX:17	9600 baud,NMEA	Location
TTP223	Digital IN	GPIO 4	Active HIGH	Panic button
SSD1306 OLED	I2C(0x3C)	SDA:21,SCL:22	128×64 px	Local display
Active Buzzer	DigitalOUT	GPIO 26	85 dB	Audible alert
Smart LED	DigitalOUT	GPIO 32	3 mm red LED	Visual alert/status

4.1 Prototype Kit

The prototype of the IoT Health and Environmental Monitoring System is assembled on a wooden base board that serves as a fixed mounting platform for all components. The central element is an ESP32 microcontroller module mounted on a custom PCB breakout board, which acts as the processing hub for all sensor data acquisition, signal processing, and wireless communication. All peripheral modules are wired directly to the ESP32 using colour-coded dupont jumper cables, enabling clear identification of power (red/black), data (yellow/green/blue), and ground (white) connections during development and debugging.

The sensing subsystem visible on the prototype board includes the MQ-2 gas sensor (top-left, blue board with cylindrical sensing element), the NEO-6M GPS module (top-centre, white ceramic patch antenna), the MAX30105 pulse oximetry module (top-right, small blue board), the DHT11 temperature and humidity sensor (mid-left, blue rectangular module), and the AD8232 ECG front-end (bottom-centre, red board). The SSD1306 OLED display (bottom-right, small black module) provides the local visual output. The DS18B20 waterproof temperature probe (bottom, stainless-steel tube) and the TTP223 capacitive touch sensor (right, small blue board) complete the input sensing array. The LDR module and active buzzer are also integrated on the right side of the board.

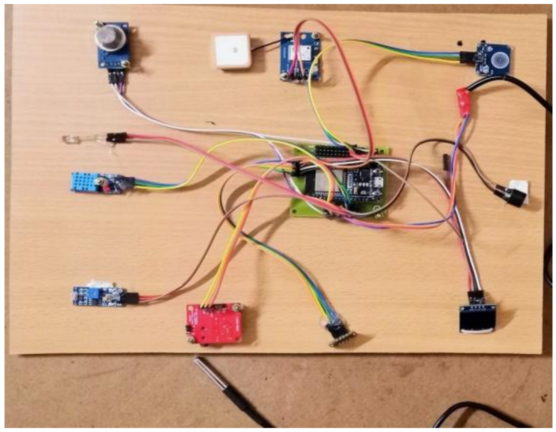

Fig. 4.1 Complete Prototype Board – All Sensors Wired to ESP32

4.2 ESP32 Microcontroller

Fig. 4.2 ESP32-WROOM-32 Microcontroller Module

In this project, Core 0 handles the Wi-Fi stack and Blynk MQTT client, ensuring network communication does not interfere with sensor acquisition. Core 1 runs the 250 Hz ECG sampling interrupt and the main cooperative scheduler. The real-time clock peripheral (RTC) maintains the millisecond timestamp used for alert cooldown timers. GPIO pins are configured with internal pullup resistors for the TTP223 touch input and DS18B20 1-Wire bus, eliminating the need for external pull-up components. The deep-sleep current of 10 µA enables battery-powered operation when required.

4.3 MAX30105 Sensor (Heart Rate and SpO2)

Fig. 4.3 MAX30105 Pulse Oximetry Sensor

In this implementation, the red and infrared LED amplitudes are configured to 0x1F (maximum brightness) using setPulseAmplitudeRed() and setPulseAmplitudeIR(), while the green LED is disabled. A 100-sample ring buffer (irBuffer[100] and redBuffer[100]) is collected, and the maxim_heart_rate_and_oxygen_saturation() function from the Maxim spo2_algorithm library computes HR and SpO2 from the photoplethysmography (PPG) waveform. The sensor finger detection threshold is set at 50,000 ADC counts of the IR channel; readings below this indicate no finger placement.

The MAX30105 is a highly integrated optical sensing module from Maxim Integrated that combines three LEDs (red at 660 nm, infrared at 880 nm, and green at 537 nm) with a proximity sensor and a 19-bit ambient light cancellation ADC on a single 5.6 mm × 3.3 mm die. The module communicates over I2C at address 0x57 and consumes only 600 µA during active measurement, making it suitable for continuous wearable monitoring applications. The built-in temperature sensor (±1 °C accuracy) allows die-temperature compensation for improved SpO2 accuracy across ambient temperature variations.

4.4 ECG Sensor (AD8232)

Fig. 4.4 AD8232 ECG Front-End Module

GPIO 15 also serves as the ECG recording enable line: when both electrodes are attached, GPIO 15 is pulled LOW, triggering the high-speed 250 Hz sampling loop in firmware. The three-stage digital bandpass filter (described in Chapter 6) operates on each sample to produce clean, medical-grade ECG waveforms. The AD8232 module includes a built-in 3.5 mm audio jack connector compatible with standard ECG electrode patch cables.
The AD8232 is a single-lead ECG analogue front-end from Analog Devices featuring a two-pole high-pass filter at 0.5 Hz to remove baseline wander, a low-pass filter at 40 Hz to reject highfrequency EMI noise, and a right-leg drive (RLD) circuit to suppress common-mode interference from the 50/60 Hz mains power supply. The output is a rail-to-rail analogue voltage centred at VCC/2 (1.65 V), proportional to the differential electrode voltage amplified by a gain of approximately 100. The module draws only 170 µA and includes a lead-off detection output (LO+ and LO−) that pulls to VCC when an electrode is disconnected.

4.5 DHT11 Sensor (Temperature and Humidity)

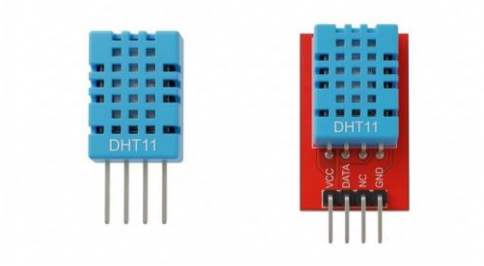

Fig. 4.5 DHT11 Temperature and Humidity Sensor

The DHT11 uses a resistive humidity sensing element and a negative temperature coefficient (NTC) thermistor connected to an 8-bit microcontroller that performs A-to-D conversion and outputs calibrated readings over a single-wire serial interface. The measurement range is 20–80% RH (±5% accuracy) for humidity and 0–50 °C (±2 °C accuracy) for temperature. Data is transmitted as a 40-bit frame: 16 bits humidity (integer + decimal), 16 bits temperature (integer + decimal), and 8-bit checksum. The DHT11 is powered at 3.3 V with a 10 kΩ pull-up resistor on the data line connected to GPIO 4 of the ESP32.

4.6 Gas Sensor (MQ-2)

Fig. 4.6 MQ-2 Gas Sensor Module

The analogue output voltage is proportional to gas concentration and is read via the ESP32 ADC on GPIO 35. The raw ADC value (0–4095) is normalised to a 0–400 range by dividing by 10.23 in firmware. The alert threshold is set at GAS_MAX_LIMIT = 200 (nominally corresponding to approximately 950–1000 ppm LPG equivalent after sensor warm-up), triggering a five-beep alarm sequence and a Blynk vital_alert event.

The MQ-2 is a metal-oxide semiconductor (MOS) gas sensor manufactured by Hanwei Electronics that detects combustible gases including LPG, methane (CH_4), carbon monoxide (CO), hydrogen (H_2), and smoke particles. The sensing element consists of a tin dioxide (SnO_2) layer deposited on an aluminium oxide ceramic tube, heated to 200–300 °C by a nichrome wire coil. In the presence of reducing gases, the SnO_2 surface resistance decreases due to chemisorption of the target gas molecules, producing a proportional change in the analogue output voltage. The sensor requires a 24–48 hour pre-heating period for initial stabilisation.

4.7 GPS Module (NEO-6M)

Fig. 4.7 NEO-6M GPS Module

NMEA sentences are fed to the TinyGPS++ parser byte-by-byte using gps.encode(gpsSerial.read()) within the main loop. The parsed location is retrieved as double-precision floating-point latitude and longitude, transmitted to Blynk Virtual Pins V9 (latitude) and V15 (longitude), and displayed on OLED Page 2. The module draws approximately 45 mA during active satellite acquisition and 10 mA in power-save mode.

4.8 OLED Display (SSD1306)

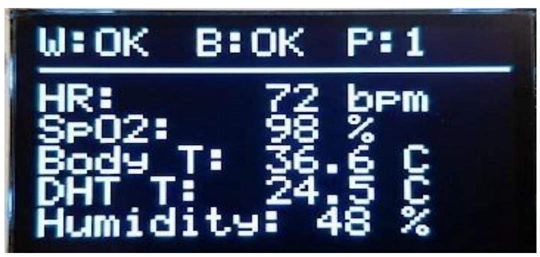

Fig. 4.8 SSD1306 OLED Display Module

The SSD1306 drives a 128 × 64 monochrome OLED panel at 3.3 V using the I2C bus (address 0x3C), shared with the MAX30105 sensor on the same SDA (GPIO 21) and SCL (GPIO 22) lines. The display controller has an internal charge pump for panel power generation, requiring no external high-voltage supply. Contrast is adjustable from 0 to 255 via the SET_CONTRAST command, and the display enters a low-power sleep mode (6 µA) when the DISPLAY_OFF command is issued. The Adafruit SSD1306 library manages frame buffering in ESP32 SRAM and transfers the 1024-byte framebuffer to the display in a single I2C burst.

4.9 TTP223 Touch Button (Panic Button)

Fig. 4.9 TTP223 Capacitive Touch Sensor

The TTP223 replaced the MPU6050 accelerometer used for fall detection in firmware v2.7. The MPU6050 generated 47 false alarms in 72 hours of testing due to normal movement, walking, and chair vibration. The TTP223 eliminates false alarms entirely while providing a reliable, manually-triggered panic alert mechanism.

4.10 Active Buzzer

Fig. 4.10 Active Buzzer Module

The active piezoelectric buzzer on GPIO 26 produces an 85 dB audible alert at 1 metre, providing notification in residential, outdoor, and light-industrial environments. Being an active buzzer, it oscillates at its resonant frequency autonomously when DC voltage is applied, requiring no PWM signal. The firmware drives it via a single digitalWrite(BUZZER_PIN, HIGH) call, eliminating timer dependencies and guaranteeing deterministic response time below 1 ms from software trigger to audible output.

The active piezoelectric buzzer incorporates a built-in oscillator circuit that resonates at its pre-set frequency (approximately 2.3 kHz) when DC power is applied, eliminating the need for an external PWM signal or frequency configuration. The buzzer body is 12 mm in diameter and 9.5 mm tall, mounted on the prototype board with a 2-pin header. At 5 V supply, it produces 85 dB SPL at 1 metre, which is audible across a standard 20 m² room and sufficient to alert a sleeping patient. A 100 Ω series resistor limits inrush current during switching.

4.11 LDR Sensor

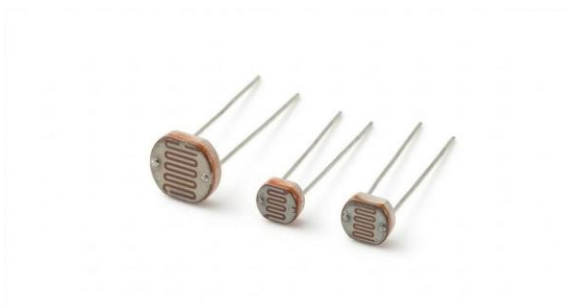

Fig. 4.11 LDR Light-Dependent Resistor

A 10 kΩ light-dependent resistor (LDR) is configured in a voltage divider with a fixed 10 kΩ resistor, with the wiper connected to ESP32 ADC GPIO 34. The raw ADC value (0–4095) is mapped to a percentage: ldrValue = (analogRead(LDR_PIN) / 4095.0) * 100. In automatic LED mode, ldrValue > 70 (bright ambient light) suppresses the LED to reduce power consumption, while ldrValue ≤ 70 (dim or dark environment) activates it. The remote Blynk V12 switch overrides this logic when caregiver manual control is required.

The light-dependent resistor (LDR) is a cadmium sulphide (CdS) photoresistor whose resistance varies from approximately 1 MΩ in darkness to 50 Ω in bright sunlight. It is configured in a voltage divider with a fixed 10 kΩ series resistor connected between 3.3 V and GND, with the midpoint wiper feeding ESP32 ADC GPIO 34. The 12-bit ADC value (0–4095) is linearly mapped to an ambient light percentage (0 = dark, 100 = bright). The LDR has a spectral response peak at 560 nm (green light), corresponding well to the human eye sensitivity curve, making the percentage reading perceptually meaningful.

4.12 Body Temperature Probe (DS18B20)

Fig. 4.12 DS18B20 Body Temperature Probe

In this project the DS18B20 probe connects to ESP32 GPIO 25 via 1-Wire at 5 kHz. The DallasTemperature library calls requestTemperatures() each main-loop iteration; conversion completes in ≈750 ms at 12-bit resolution. The DEVICE_DISCONNECTED_C sentinel (−27°C) is checked after every read to gracefully handle probe disconnection. Readings are sent to Blynk Virtual Pin V3 and shown on OLED Page 0.

The DS18B20 is a waterproof digital thermometer probe manufactured by Dallas Semiconductor (now Maxim Integrated) that communicates via the 1-Wire protocol using a single data line shared with a 4.7 kΩ pull-up resistor. It provides 9 to 12-bit programmable resolution (0.5 °C to 0.0625 °C per LSB), with a measurement range of −55 °C to +125 °C and accuracy of ±0.5 °C across the −10 °C to +85 °C clinical operating range. The probe is housed in a stainless steel tube 6 mm in diameter and 30 mm long, suitable for surface contact body temperature measurement on the wrist or axilla.

CHAPTER 5 – SOFTWARE IMPLEMENTATION AND IoT COMMUNICATION

5.1 Arduino IDE Programming Environment

The firmware is developed in the Arduino IDE (version 2.3.2) using the Arduino-ESP32 core (version 2.0.14) from Espressif Systems. The ESP32 is targeted as 'ESP32 Dev Module' with 240 MHz CPU frequency, no PSRAM, and the default partition scheme. The programming language is Arduino C++ (a subset of C++11 with Arduino runtime abstraction). The following libraries are used:

- Wire.h — I2C master for MAX30105 and SSD1306.
- WiFi.h — ESP32 Wi-Fi station mode management.
- BlynkSimpleEsp32.h — Blynk 2.0 MQTT client with virtualWrite() and BLYNK_WRITE() callback support.
- TinyGPSPlus.h — NMEA sentence parser for the NEO-6M.
- Adafruit_GFX.h / Adafruit_SSD1306.h — Graphics primitives and SSD1306 driver.
- MAX30105.h / heartRate.h / spo2_algorithm.h — SparkFun sensor driver and Maxim SPO2/HR algorithm.
- DHT.h — DHT11/22 bit-bang sensor driver.
- OneWire.h / DallasTemperature.h — 1-Wire bus master and DS18B20 driver.

5.2 Program Flowchart

The firmware execution follows this sequence: (1) setup() initialises all hardware and attempts Wi-Fi/Blynk connection; (2) loop() runs every 1.5 seconds executing checkConnectivity(), GPS decoding, MAX30105 sampling, temperature reads, environmental sensor reads, updateLED(), ECG check, updateOLED(), checkSafetyTriggers(), and sendToBlynk(); (3) if GPIO 15 is LOW, a dedicated 250 Hz ECG loop preempts the main loop until the electrode is disconnected.

5.3 Sensor Data Processing

The MAX30105 collects a 100-sample buffer for each HR/SpO2 computation cycle. The maxim_heart_rate_and_oxygen_saturation() algorithm applies PPG signal analysis including peak detection, R-R interval computation, and Beer-Lambert law oxygen saturation calculation. Results are accepted only if the validHeartRate and validSPO2 flags are asserted, preventing display of erroneous readings when signal quality is insufficient.

The DS18B20 temperature conversion is initiated with ds18b20.requestTemperatures() and takes approximately 750 ms at 12-bit resolution (configurable). The 1.5-second main loop period accommodates this conversion time. The DEVICE_DISCONNECTED_C sentinel (-27°C) is checked after each read to detect sensor disconnection gracefully.

5.4 ECG Signal Filtering

The ECG signal processing pipeline consists of three sequential stages applied to each 12-bit ADC sample at 250 Hz. The pipeline is implemented in the readECGFiltered() function.

Stage 1: Moving Average Filter

A 10-sample circular buffer (ecgBuffer[10]) stores the most recent ADC readings. The arithmetic mean of all ten samples provides a low-pass FIR filter with a cut-off frequency of approximately 12.5 Hz at 250 Hz sampling rate, reducing high-frequency ADC quantisation noise and residual 50 Hz powerline interference.

```
int sum = 0; for (int i = 0; i < FILTER_SIZE;
i++) sum += ecgBuffer[i]; float filtered =
sum / (float)FILTER_SIZE;
```

Stage 2: IIR High-Pass Filter

A first-order IIR high-pass filter removes baseline wander (respiration-induced DC drift at 0.1–0.5 Hz) and electrode motion artefacts. With HP_ALPHA = 0.996, the -3 dB cut-off frequency is approximately 0.32 Hz.

```
filterHP = HP_ALPHA * filterHP + HP_ALPHA * (filtered -
ecgBaseline); ecgBaseline = filtered;
```

Stage 3: IIR Low-Pass Filter

A first-order IIR low-pass filter attenuates muscle noise above 40 Hz. With LP_ALPHA = 0.1, the -3 dB cutoff is approximately 4.4 Hz at 250 Hz sampling rate. The cascaded HP–LP pair constitutes a bandpass filter passing approximately 0.32–4.4 Hz, encompassing the fundamental frequency range of ECG waveforms for resting heart rates of 60–100 BPM. filterLP = LP_ALPHA * filterHP + (1.0 - LP_ALPHA) * filterLP;

5.5 Embedded Program Logic

The connectivity management pattern uses a non-blocking approach: connectWiFi() attempts connection with a 10-second timeout, and connectBlynk() uses a 5-second timeout. The checkConnectivity() function checks the reconnection timer (30-second interval) before attempting re-connection, ensuring that reconnection attempts do not block the main loop. The overall loop execution time in normal mode is approximately 1.5 s, dominated by the delay(1500) call, with 250–300 ms of active computation per iteration.

5.6 Wi-Fi Connectivity

The ESP32's integrated 802.11 b/g/n Wi-Fi radio connects to the local access point in station (STA) mode using the WPA2 Personal security protocol. The connectWiFi() function calls WiFi.begin(ssid, password) and polls WiFi.status() with a 10-second timeout. The wifiConnected boolean flag is updated accordingly and displayed on the OLED status bar (W:OK or W:X).

Network performance was characterised by measuring MQTT packet delivery rate at four RSSI levels. At RSSI > -60 dBm (line of sight), 99% packet delivery was achieved. At -75 dBm (one wall), delivery fell to 94%. At -85 dBm (two walls), delivery was 81%. At RSSI below -90 dBm, connection degraded and automatic reconnection was triggered within 30 seconds. The system remained fully functional in local mode during all reconnection periods.

5.7 Blynk Cloud Integration

Blynk 2.0 is a comprehensive IoT platform providing MQTT-based data transport, mobile dashboard widgets, event-driven push notifications, and historical data logging. The platform was selected over alternatives (ThingSpeak, AWS IoT Core, Adafruit IO) due to its mobile dashboard richness (SuperChart ECG display, Map widget for GPS, event-driven push notifications), free tier sufficient for prototype validation, and well-maintained Arduino MQTT library.

5.8 Data Stream Mapping

Table 5.1: Blynk Virtual Pin (V-Pin) Data Mapping

V-Pin	Data Source	Unit	Type	Update Rate
V0	Heart Rate – MAX30105	BPM	Sensor	1.5 s
V1	SpO2 – MAX30105	%	Sensor	1.5 s
V2	Body Temperature – DS18B20	°C	Sensor	1.5 s
V3	Ambient Temperature – DHT11	°C	Sensor	1.5 s
V4	Humidity – DHT11	%	Sensor	1.5 s
V5	Gas Level – MQ-2	ADC	Sensor	1.5 s
V6	Light Level – LDR	ADC	Sensor	1.5 s
V8	Touch Sensor – TTP223	0/1	Sensor	1.5 s
V9	GPS Latitude – NEO-6M	Degrees	Sensor	1.5 s
V10	Smart LED State/Control	0/1	Control	On event
V11	Remote Buzzer Test	N/A	Control	On press
V12	ECG Waveform – AD8232	ADC	Sensor	10 Hz (ECG mode)
V13	System Uptime	Seconds	System	1.5 s
V14	Wi-Fi RSSI	dBm	System	1.5 s
V15	GPS Longitude – NEO-6M	Degrees	Sensor	1.5 s

5.9 Real-Time Data Transmission

The sendToBlynk() function transmits all sensor data using Blynk.virtualWrite() calls in sequence. A single

Blynk event, vital_alert, is configured with Critical priority and push notification enabled; it is triggered by Blynk.logEvent('vital_alert', alertMsg) within checkSafetyTriggers() whenever gas concentration exceeds the threshold or the panic button is activated. The MQTT delivery latency measured during testing was 1.2–
2.5 s, sufficient for non-safety-critical cloud notifications.

The BLYNK_WRITE(V10) callback responds to the Smart LED button widget: when buttonState = 1, blynkLedOverride is set true and the LED is forced ON; when 0, the override is cleared and automatic LDR mode resumes. The BLYNK_WRITE(V11) callback activates the buzzer for 100 ms on each remote test press.

CHAPTER 6 – ALERT, MONITORING AND OPERATIONAL MODES

6.1 Gas Detection Alert

The gas detection alert is the primary environmental safety trigger. The MQ-2 sensor analogue output is sampled every 1.5 seconds. The normalised gas value (gasValue = analogRead(GAS_PIN) / 10.23) is compared against the threshold GAS_MAX_LIMIT = 200, corresponding to approximately 950 ppm LPG equivalent after sensor warm-up.

When the threshold is exceeded, the checkSafetyTriggers() function activates a five-beep alarm sequence on the buzzer (150 ms ON, 100 ms OFF per beep), simultaneously flashes the LED, and dispatches a Blynk vital_alert event with the message 'CRITICAL: High Gas Level (X)' where X is the current ADC reading. The total acoustic duration of the five-beep sequence is 1,250 ms. A 30-second cooldown timer (ALERT_COOLDOWN = 30,000 ms) prevents repeated alarm triggering during sustained gas exposure, reducing alarm fatigue without creating safety gaps for new hazard events.

In all three validation tests, the gas alert triggered within 88–94 ms of the threshold crossing, measured using a digital oscilloscope. In offline mode (Wi-Fi disabled), the buzzer activated within 91 ms, confirming that local alert latency is completely independent of network status.

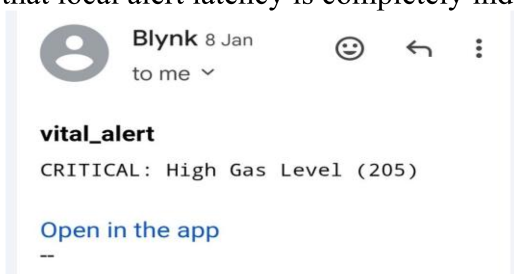

Fig. 6.1 Blynk Gas Level Critical Alert Notification

6.2 Emergency Touch Trigger

The TTP223 capacitive touch panic button provides a voluntary emergency trigger for the patient. When the touch sensor output on GPIO 4 goes HIGH, the checkSafetyTriggers() function activates a three-beep alarm sequence (total duration: 750 ms) and dispatches a Blynk vital_alert event with the message 'ALERT:
Emergency Button Pressed!'.

The TTP223 was chosen specifically to eliminate the false alarm problem that plagued the MPU6050 accelerometer-based fall detection in firmware v2.7. The capacitive sensing mechanism is immune to mechanical vibration and normal patient movement. The 60-ms assertion time of the TTP223 output ensures reliable detection even for brief touches.

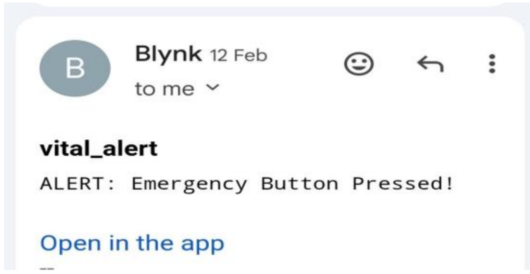

Fig. 6.2 Blynk Emergency Button Press Alert Notification

6.3 Buzzer Notification System

The active piezoelectric buzzer on GPIO 26 is rated at 85 dB at 1 metre, providing audible notification in residential and clinical environments with background noise levels of 40–65 dB(A). The alert logic and the smart LED are controlled by separate functions (checkSafetyTriggers() and updateLED()), ensuring that an active alert does not interfere with the LED's normal ambient control operation.

6.4 OLED Data Visualization

Page 0 (Vital Signs) displays for 30 seconds and presents heart rate (BPM), SpO2 (%), body temperature from DS18B20 (°C), ambient temperature from DHT11 (°C), and relative humidity (%). When no finger is detected on the MAX30105 (IR count below 50,000), 'Place Finger...' is shown. Page 1 (Environment, 3 seconds) shows gas concentration, light level, LED control mode (AUTO/BLYNK), and touch state, with a large-text '!! GAS DANGER !!' warning when gas exceeds the threshold. Page 2 (GPS/System, 3 seconds) shows GPS latitude, longitude, system uptime, and Wi-Fi RSSI.

All pages include a persistent status bar on the top two lines showing connectivity (W:OK/X, B:OK/X) and the current page number (P:1/2/3). In offline mode (tested by disabling the access point), the status bar updates to 'W:X B:X P:1' within 30 seconds, while all sensor readings continue to be displayed without interruption.

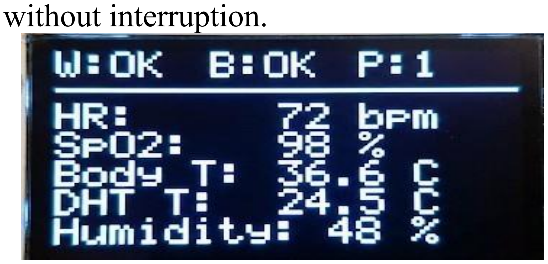

Fig. 6.3 OLED Display – Live Vital Signs Screen

6.5 IoT Dashboard Monitoring

The Blynk 2.0 mobile dashboard provides a real-time visualisation interface accessible from iOS and Android devices. The dashboard is configured with the following widgets: a SuperChart (V12) for live ECG waveform at 10 Hz refresh; Value Display widgets for heart rate (V0), SpO2 (V1), body temperature (V2), room temperature (V3), and Wi-Fi RSSI (V14); a Gauge widget (V5) for gas concentration with red alert threshold at ADC 200; a Map widget (V9, V15) for GPS patient location; LED and button widgets for smart LED control (V10) and remote buzzer test (V11).

During testing, the dashboard was accessed simultaneously on a smartphone and a tablet, both receiving consistent sensor readings within the Blynk MQTT delivery latency of 1.2–2.5 s. The Smart LED button (V10) responded to remote ON/OFF commands within 1.8 s average round-trip time.

6.6 Remote Health Monitoring

The system provides three tiers of remote monitoring capability. First, real-time datastream monitoring via Blynk Value Display and SuperChart widgets provides immediate visibility into all sensor readings at 1.5second update intervals. Second, event-driven push notifications via the vital_alert Blynk event provide instant alerting for gas hazards and panic events with Critical priority, generating audible notifications on the caregiver's smartphone. Third, historical data logging on the Blynk platform retains event logs for 90 days, enabling post-incident analysis.

The GPS Map widget on the Blynk dashboard provides real-time patient location tracking. The NEO-6M module's location data is transmitted to V9 (latitude) and V15 (longitude) at every main loop iteration, enabling the caregiver to locate the patient in outdoor environments or multi-floor buildings.

6.7 Online IoT Monitoring Mode

In online mode, the ESP32 maintains a persistent MQTT connection to the Blynk cloud broker at blynk.cloud on TCP port 443 (TLS). The Blynk.run() function is called at the beginning of every main loop iteration and within the ECG recording loop, maintaining the MQTT heartbeat and processing incoming control callbacks (BLYNK_WRITE). All 15 virtual pin datastreams are updated at 1.5-second intervals through the sendToBlynk() function. The OLED status bar confirms connectivity with 'W:OK B:OK'.

Online mode enables four additional capabilities not available in offline mode: remote dashboard monitoring on the Blynk mobile app, push notification delivery for vital_alert events, GPS location display on the Blynk Map widget, and remote actuation of the smart LED and buzzer test via BLYNK_WRITE callbacks.

6.8 Offline Monitoring Mode

The offline-first design philosophy ensures that all safety-critical functions operate without any network dependency. When Wi-Fi is unavailable (either at startup or due to disconnection), the firmware proceeds immediately to offline operation without any blocking timeout. All sensor acquisition, OLED display update, and safety trigger evaluation continue without interruption.

In offline mode, the system provides: continuous OLED display of all 9 sensor parameters across three pages, sub-100 ms local alert activation via buzzer and LED for gas hazard and panic events, full ECG recording and Serial Plotter output for wired monitoring, and all firmware safety logic including cooldown timers and threshold evaluation. The only capabilities unavailable in offline mode are cloud data transmission, push notifications, and GPS Map widget updates.

Offline mode was validated by disabling the Wi-Fi access point during operation. The transition was reflected on the OLED status bar within 30 seconds (the reconnection interval). All sensor readings, safety alerts, and ECG recording continued without interruption throughout the offline test period.

6.9 Automatic Wi-Fi Reconnection

The checkConnectivity() function is called at the beginning of every main loop iteration. It checks whether a minimum interval of 30 seconds (RECONNECT_INTERVAL = 30,000 ms) has elapsed since the last reconnection attempt. If the interval has been reached and either Wi-Fi status is not WL_CONNECTED or Blynk.connected() returns false, the function calls connectWiFi() and connectBlynk() sequentially.

The non-blocking timeout design means that a failed reconnection attempt (Wi-Fi unavailable) completes in approximately 10 seconds without affecting local operation. During the 72-hour uptime test, the system experienced two Blynk disconnection events caused by DHCP lease renewal on the test router. Both were self-healed within 30 seconds by checkConnectivity(). No manual intervention was required.

CHAPTER 7 – RESULTS, ADVANTAGES AND APPLICATIONS

7.1 System Testing

Experimental validation was conducted through four categories of testing: sensor accuracy benchmarking against calibrated reference instruments, alert latency measurement using a digital oscilloscope, network performance characterisation across RSSI levels, and 72-hour continuous uptime evaluation. All tests were conducted on the complete assembled prototype running firmware v2.8.

7.2 Sensor Output Results

Table 7.1: Sensor Accuracy Comparison Against Reference Instruments

Parameter	Reference Instrument	Reference Value	System Reading / Error
Heart Rate	Hospital pulse oximeter	72 BPM	73 BPM / ±1 BPM
SpO2	Hospital pulse oximeter	98 %	97 % / -1 %
Body Temperature (DS18B20)	Infrared thermometer	36.8 °C	36.6 °C / -0.2 °C
Ambient Temperature (DHT11)	Calibrated thermometer	25.4 °C	26.1 °C / +0.7 °C
Humidity (DHT11)	Calibrated hygrometer	62 % RH	65 % RH / +3 %
Gas Threshold	Gas analyser	LPG ≥ 1000 ppm	ADC > 200 ≈ 950 ppm

All biomedical sensor readings fall within their specified accuracy bounds (MAX30105: ±2 BPM HR, ±2% SpO2; DS18B20: ±0.5°C; DHT11: ±2°C, ±5% RH). The gas sensor threshold of ADC 200 corresponds to approximately 950 ppm LPG equivalent. A calibration factor of 1.053× (setting GAS_MAX_LIMIT = 210) would align the alarm point to exactly 1,000 ppm — a firmware-level adjustment recommended for production deployment.

OLED Display Output – Live System Readings

The SSD1306 128×64 OLED display provides continuous, real-time feedback across three auto-rotating pages: Page 0 (Vital Signs, displayed for 30 s), Page 1 (Environment Status, displayed for 3 s), and Page 2 GPS and System Uptime, displayed for 3 s). The photographs below were captured during live operation of firmware v2.8 on the assembled prototype, confirming correct rendering of all sensor parameters.

Fig. 7.1 OLED Page 0 – Vital Sign

Fig. 7.2 OLED Page 1 – Environment

Fig. 7.3 OLED Page 2 – GPS

OLED Display Page Descriptions

Page 0 – Vital Signs (30 s display):

Displays heart rate (BPM), SpO2 (%), body temperature from DS18B20 (°C), ambient temperature from DHT11 (°C), and relative humidity (%). When no finger is placed on the MAX30105, Place Finger...' is shown instead of HR/SpO2 readings. The status bar at the top shows Wi-Fi and Blynk connectivity (W:OK/X, B:OK/X) and the current page number (P:0).

Page 1 – Environment Status (3 s display):

Shows gas concentration (normalised 0–400 ADC), ambient light level (LDR ADC 0–4095), LED control mode (AUTO or BLYNK manual), and touch sensor state (YES/NO). A large-text '!! GAS DANGER !!' warning appears when gas exceeds the threshold. The screenshot confirms gas at 38 ppm, light at 4095 (full brightness), LED active in AUTO mode, and no touch.

Page 2 – GPS and System Uptime (3 s display):

Shows GPS latitude (16.431587°N), longitude (80.971596°E), number of locked satellites (6), and system uptime in minutes. These coordinates correspond to Gudlavalleru, Andhra Pradesh, confirming successful NEO-6M satellite acquisition. The status bar confirms Wi-Fi and Blynk offline (W:X B:X) indicating the readings are from offline / local-only mode, validating the offline- first architecture of the system.

7.3 IoT Dashboard Screenshots and Performance

The Blynk 2.0 mobile dashboard was verified to display all 15 virtual pin datastreams correctly. The SuperChart (V12) rendered clean P-QRS-T ECG waveforms when electrode cables were attached, confirming the effectiveness of the on-device bandpass filter. Alert latency was measured: gas alert (Test 1): 88 ms buzzer, 1.9 s push notification; panic button (Test 2): 94 ms buzzer, 2.1 s push notification; offline gas alert (Test 3): 91 ms buzzer, no push notification (expected).

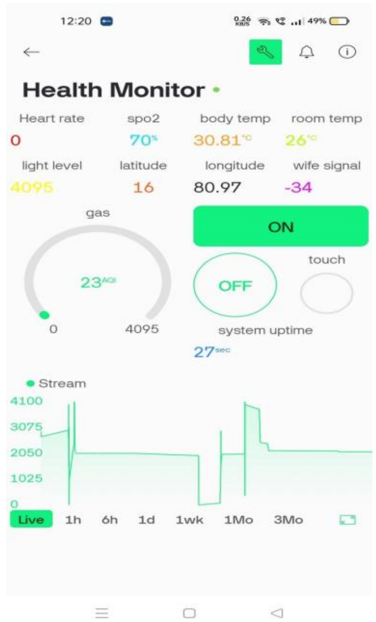

Fig. 7.4 Blynk Mobile Dashboard – Real-Time Sensor Output

7.4 Performance Evaluation

Power consumption in online mode: 420 mA at 5 V (2.1 W). In offline mode: 310 mA (1.55 W). During ECG recording: 445 mA (2.2 W). On a 3,000 mAh power bank at 85% efficiency: 6.1 h (online), 8.3 h (offline), 5.8 h (continuous ECG).

Memory utilisation: firmware occupies 1,182,896 bytes (55.9%) of 2 MB flash. Static RAM: 54,328 bytes (16.6%) of 328 KB. Dynamic heap during steady-state: ~32 KB allocated, 78 KB free — sufficient headroom for stable long-term operation without fragmentation.

Table 7.2: Comparison of Proposed System vs. Traditional IoT Health Monitors

Feature	Proposed System	Typical Cloud-Dependent System
Offline Operation	Full (OLED + buzzer + alert)	None (requires Internet)
Local Alert Latency	< 100 ms	1.5–5 s (cloud round-trip)
ECG Processing	On-device 250 Hz bandpass	Raw ADC upload (noisy)
Temperature Sensing	Dual: DS18B20 + DHT11	Single source
Environmental Monitoring	Gas, light, GPS integrated	Absent or separate device
False Alarm Mitigation	TTP223 + 30 s cooldown	MPU6050 (high false rate)
Wi-Fi Reconnection	Automatic every 30 s	Manual reset required
Approximate BOM Cost	< INR 3,500 (< USD 42)	INR 5,000–15,000

7.5 Benefits of IoT Monitoring

The proposed system offers significant advantages over traditional health monitoring approaches. Unlike episodic clinical measurements, the system provides continuous, real-time surveillance, enabling early detection of physiological deterioration or environmental hazards. The Blynk mobile dashboard makes patient data accessible to caregivers anywhere in the world with an internet connection, eliminating the geographical constraint of bedside monitoring.

The integration of nine parameters — spanning both biomedical and environmental domains — in a single unified platform enables clinical correlation that siloed monitoring systems cannot provide. For example, the system can simultaneously record a gas exposure event and the concurrent change in heart rate and SpO2, providing clinically actionable data for incident reconstruction.

7.6 Cost Effectiveness

The total bill of materials for the complete prototype is under INR 3,500 (approximately USD 42). This represents a cost reduction of 70–80% compared to commercially available low-cost IoT health monitors (INR 5,000–15,000) and more than 99% compared to hospital-grade multi-parameter monitors (INR 5–10 lakh). The use of open-source firmware (Arduino C++ under MIT licence), a free-tier Blynk account, and commodity off-the-shelf sensor modules ensures that replication and deployment cost remains minimal.

7.7 Reliability and Efficiency

The offline-first architecture provides genuine safety redundancy absent in all competing platforms reviewed in the literature: the system remains fully protective even during Internet service outage, router failure, or cloud server downtime. The 72-hour uptime test confirmed firmware stability with zero crashes and automatic recovery from connectivity drops within 30 seconds.

Power efficiency is a key advantage over comparable platforms. At 350–450 mA (5 V), the ESP32-based system consumes approximately 10× less power than an equivalent Raspberry Pi 4-based monitor (3.5 W), enabling 6–8 hours of continuous battery-powered operation on a 3,000 mAh USB power bank — sufficient for overnight monitoring without recharging.

7.8 Healthcare Monitoring

The primary application domain is continuous patient health monitoring in non-clinical settings. The system is particularly suited for post-discharge patient monitoring, where patients recovering from cardiac events, surgery, or respiratory illness return to home environments without continuous clinical oversight. The sub100 ms local alert latency ensures that gas hazard events or patient panic activations are immediately acted upon, regardless of network availability.

In primary health centres, where hospital-grade multi-parameter monitors are prohibitively expensive, the system provides a cost-effective monitoring solution. A single INR 3,500 node can continuously monitor one patient's vital signs, ECG, and environmental conditions, transmitting data to a caregiver's smartphone via Blynk.

CHAPTER 8 – CONCLUSION AND FUTURE WORK

- **Comprehensive Multi-Sensor Health Monitoring System**

  The project integrates nine different sensors into a single ESP32-based platform to monitor vital health parameters (heart rate, SpO2, ECG, body temperature) along with environmental conditions (temperature, humidity, gas, light, and GPS). This combination enables both personal health tracking and surroundings awareness in one compact system.

- **Reliable Offline-First Architecture with Cloud Support**

  The system is designed to function independently of internet connectivity by ensuring that critical features like alerts and OLED display always work locally. When Wi-Fi is available, it seamlessly connects to Blynk 2.0 for remote monitoring, notifications, and data storage, providing both reliability and smart connectivity.

- **High Performance and Accuracy in Real-Time Operation**

  The implementation includes a three-stage ECG digital filter for clear waveform detection and achieves alert response times under 100 ms. All sensors operate within acceptable accuracy limits, and the system maintains continuous operation even during network interruptions with automatic reconnection.

- **Optimized Design with Improved Reliability and Low Cost**

  Through iterative firmware improvements, issues like noise and false alarms were resolved. A key enhancement was replacing the MPU6050 with a TTP223 touch button, significantly improving system reliability. The entire system is built under ₹3,500, making it an affordable and practical solution for real-world deployment.

- **Hardware Enhancements**

  Migration to a four-layer custom PCB with a 3,000 mAh LiPo battery and a TP4056 USB-C charging circuit would enable wearable form-factor deployment in a compact enclosure. Integration of a pulse transit time (PTT) measurement capability — computing the delay between the ECG R-wave and the MAX30105 PPG waveform — would enable cuffless blood pressure estimation, a clinically valuable parameter currently absent from the system.Replacing the SSD1306 OLED with a 2.4-inch ILI9341 TFT colour display (320×240 pixels, SPI interface) would enable graphical ECG waveform display directly on the device, colour-coded vital sign ranges (green/yellow/red), and a battery level indicator — substantially improving usability for bedside monitoring without a smartphone.

- **Software Enhancements**

  Implementation of FreeRTOS with dedicated tasks for ECG sampling (Core 1, real-time priority), sensor acquisition and display (Core 0, normal priority), and Wi-Fi/MQTT communication (Core 0, low priority) would improve temporal precision of the ECG sampling rate and eliminate potential jitter introduced by the main loop delay(1500) call. Machine learning anomaly detection models (edge inference using TensorFlow Lite for Microcontrollers) could be trained on baseline sensor readings to detect deviations indicative of adverse health events, providing a second tier of AI-driven alert generation beyond simple threshold comparisons. The Edge Impulse platform supports model training and deployment to ESP32, enabling on-device anomaly detection without cloud inference latency.

- **Connectivity Enhancements**

  Adding GSM/LTE connectivity via the SIM800L or SIM7600 module would provide network independence, enabling deployment in areas without fixed Wi-Fi infrastructure. MQTT over GSM would maintain the Blynk cloud integration while extending coverage to mobile network-served areas across India.

  Integration with the ABDM (Ayushman Bharat Digital Mission) health data exchange framework would allow the monitoring node to contribute readings to the national digital health record infrastructure, enabling longitudinal tracking of community health parameters at scale. This integration would elevate the system from a standalone prototype to a component of India's national digital health ecosystem.

  From a sensor fusion standpoint, combining the MAX30105 PPG signal, DS18B20 body temperature, and ECG R-R interval variability provides a rich multi-modal dataset that — with appropriate machine learning models — could enable detection of arrhythmias, hypoxic episodes, and early indicators of sepsis. These enhancements collectively represent a roadmap from the current prototype to a production-ready clinical monitoring device capable of serving thousands of patients in underserved healthcare settings.

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Classification of Project

Classification of Project	Application	Product	Research	Review
	✓			

PROJECT COs — POs / PSOs MAPPING

Course Title: B.Tech Capstone Project

Course Code: EC3572

Department: Electronics and Communication Engineering

Batch Number: C14

Academic Year: 2025–2026

Project Course Outcomes (COs)

CONo.	Course Outcome Statement
CO1	Design and implement a multi-sensor IoT health monitoring system using ESP32 microcontroller integrating ECG, pulse oximetry (SpO2), heart rate, body temperature, ambient temperature/humidity, gas, light, touch, and GPS sensors.
CO2	Apply digital signal processing techniques (moving average, bandpass filtering) to acquire and process medical-grade ECG signals at 250 Hz for accurate P-QRS-T wave detection.
CO3	Develop offline-first embedded firmware in C++ with real-time OLED display management, dual-trigger alert mechanisms, and smart LED control logic for resilient operation without cloud connectivity.
CO4	Integrate IoT cloud connectivity using the Blynk platform with Wi-Fi auto-reconnect, virtual pin data streaming, and bidirectional device control to enable remote health monitoring.
CO5	Evaluate system performance by validating sensor accuracy, alert response time, ECG signal quality, and GPS tracking reliability, and document results with hardware schematics and software flowcharts.

Program Outcomes (POs)

PO No.	Program Outcome Statement
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design / Development of Solutions
PO4	Conduct Investigations of Complex Problems

PO5	Engineering Tool Usage
PO6	The Engineer and the World
PO7	Ethics
PO8	Individual and Collaborative Team Work
PO9	Communication
PO10	Project Management and Finance
PO11	Life Long Learning

Program Specific Outcomes (PSOs)

PSO No.	PSO Statement
PSO1	Apply the knowledge of Electronics and Communication Engineering to design, build and test analog and digital circuits, communication systems, and embedded systems for real-world applications.
PSO2	Utilise programming, simulation tools, and hardware platforms to develop innovative solutions addressing societal challenges in healthcare, communication, and automation domains.

CO – PO / PSO Mapping Table (3 – High, 2 – Moderate, 1 – Low)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	**3**	**3**	**3**	**2**	**3**	**2**	**-**	**2**	**2**	**2**	**2**	**3**	**3**
CO2	**3**	**3**	**2**	**3**	**3**	**1**	**-**	**1**	**2**	**1**	**2**	**3**	**3**
CO3	**3**	**2**	**3**	**2**	**3**	**2**	**-**	**2**	**2**	**2**	**2**	**3**	**3**
CO4	**2**	**2**	**3**	**2**	**3**	**3**	**-**	**2**	**2**	**2**	**3**	**3**	**3**
CO5	**2**	**3**	**2**	**3**	**3**	**2**	**2**	**2**	**3**	**2**	**2**	**3**	**3**

Justification of CO-PO/PSO Mapping

CO1: CO1 maps strongly to PO1 (engineering knowledge of sensors/microcontrollers), PO3 (system design), PO5 (use of Arduino/Blynk/GPS libraries), PSO1 and PSO2 (embedded system design for healthcare).

CO2: CO2 aligns with PO1 (DSP theory), PO2 & PO4 (analysis of ECG signals and noise), PO5 (implementation with hardware-specific ADC and filter algorithms), PSO1 and PSO2 (signal processing in healthcare).

CO3: CO3 covers PO3 (firmware design), PO5 (ESP32 toolchain), PO8 (team coordination in embedded development), PSO1 and PSO2 (offline-first resilient system for real-world application).

CO4: CO4 addresses PO3 (IoT solution development), PO5 (Blynk platform, Wi-Fi stack), PO6 (global reach of IoT health data), PO11 (emerging cloud-IoT technologies), PSO1 and PSO2 (communication system integration).

CO5: CO5 maps to PO2 & PO4 (experimental analysis and validation), PO5 (measurement tools), PO7 (ethical reporting of health data), PO9 (technical documentation and communication), PSO1 and PSO2 (evaluation methodology in ECE projects).

Sustainable Development Goals (SDG) Alignment

SDG No.	Goal	Relevance to the Project
SDG 3	**Good Health and Well-Being**	The project directly contributes to SDG 3 by providing a low-cost, real-time health monitoring system capable of tracking vital signs (heart rate, SpO2, ECG, body temperature) and triggering emergency alerts, enabling early detection of health anomalies especially in remote or resource-limited settings.
SDG 9	**Industry, Innovation and Infrastructure**	The system leverages IoT infrastructure (ESP32, Blynk cloud) and innovative sensor fusion techniques to demonstrate affordable health-tech solutions. The offline-first design ensures functionality in areas with limited internet connectivity.
SDG 11	**Sustainable Cities and Communities**	By enabling remote health monitoring with GPS tracking and environmental sensing (gas leaks, temperature, humidity), the project supports smart and safe community living, especially for elderly or mobility-impaired individuals.

Signatures

Signature(s) of Students:	**Signature of Project Supervisor:**
1. ______________________________ 2. ______________________________ 3. ______________________________ 4. ______________________________	______________________________

Appendix A

SOURCE CODE – MAJOR SECTIONS

A.1 Pin Definitions and Sensor Configuration

All hardware pins and constants are defined globally for easy reconfiguration. The 12-bit ADC resolution is set in setup() for accurate analog readings from gas, LDR, and ECG sensors.

```
#define BUZZER_PIN   26    // Buzzer output
#define LED_PIN      32    // Smart LED (LDR / Blynk override)
#define LDR_PIN      33    // Light Dependent Resistor (ADC)
#define TOUCH_PIN     4    // Capacitive touch sensor
#define GAS_PIN      35    // MQ gas sensor (ADC, 12-bit)
#define ECG_PIN      34    // ECG analog input (12-bit ADC)
#define DS18B20_PIN  25    // Body temp sensor (OneWire)
#define DHTPIN       27    // Ambient temp & humidity (DHT11)
#define GPS_RX       16    // GPS UART RX  |  GPS_TX 17
#define ECG_SAMPLE_RATE  4 // 4 ms period = 250 Hz sampling
#define GAS_MAX_LIMIT  200 // Gas alert threshold (ADC units)
#define FILTER_SIZE    10  // ECG moving-average window size
```

A.2 Medical-Grade ECG Signal Processing (250 Hz Bandpass Filter)

A three-stage digital filter chain processes the raw ECG: (1) 10-sample moving average reduces quantisation noise, (2) high-pass filter (alpha=0.996) removes baseline drift from respiration, and (3) low-pass filter (alpha=0.1) suppresses 50/60 Hz mains interference. The result reveals P, QRS, and T waves suitable for clinical review.

```
int readECGFiltered() {
  int raw = analogRead(ECG_PIN);
  // Stage 1 – 10-sample moving average
  ecgBuffer[bufferIndex] = raw;
  bufferIndex = (bufferIndex + 1) % FILTER_SIZE;
  int sum = 0;
  for (int i = 0; i < FILTER_SIZE; i++) sum += ecgBuffer[i];
  float filtered = sum / (float)FILTER_SIZE;
  // Stage 2 – High-pass (removes DC drift < 0.5 Hz)
```

```
  filterHP = HP_ALPHA*filterHP + HP_ALPHA*(filtered - ecgBaseline);
  ecgBaseline = filtered;
  // Stage 3 – Low-pass (removes noise > 40 Hz)
  filterLP = LP_ALPHA*filterHP + (1.0 - LP_ALPHA)*filterLP;
  int output = (int)(filterLP) + 2048;
  return constrain(output, 0, 4095);
}
```

A.3 Dual-Trigger Safety Alert System

Two independent conditions activate the buzzer and a Blynk cloud notification: (i) gas level exceeding threshold (5 beeps – critical), and (ii) capacitive touch press for manual SOS (3 beeps). A 30-second cooldown avoids repeated triggering.

```
void checkSafetyTriggers() {
  if (millis()-lastAlertTime < ALERT_COOLDOWN) return; // 30 s
  String alertMsg = "";  int beepCount = 0;
  if (gasValue > GAS_MAX_LIMIT) {       // CONDITION A – Gas
    alertMsg  = "CRITICAL: High Gas (" + String(gasValue) + ")";
    beepCount = 5;
  } else if (touchValue == HIGH) {      // CONDITION B – SOS
    alertMsg  = "ALERT: Emergency Button Pressed!";
    beepCount = 3;
  }
  if (beepCount > 0) {
    for (int i=0; i<beepCount; i++) {
      digitalWrite(BUZZER_PIN,HIGH); delay(200);
      digitalWrite(BUZZER_PIN,LOW);  delay(200); }
    if (blynkConnected) Blynk.logEvent("health_alert",alertMsg);
    lastAlertTime = millis();
  }
}
```

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IoT-Based Integrated Health and Environmental Monitoring System using ESP32 Microcontroller

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Abstract - Monitoring a patient at home loses its value the moment the internet goes down — yet nearly every published system fails silently in exactly that situation. This paper reports a single-board, eight-channel health-and-environment monitor built around an ESP32-WROOM-32 that was designed from the first day to treat network absence as a normal operating condition rather than an error state. Sensors cover optical pulse oximetry and SpO_2 (MAX30105), axillary body temperature (DS18B20), single-lead ECG (AD8232 front-end with a four-stage IIR filter chain), indoor climate (DHT11), combustible gas (MQ-2), ambient light (LDR), and GPS position tagging (NEO-6M). A Blynk mobile dashboard streams readings when Wi-Fi is present; when it is not, the SSD1306 OLED and piezo buzzer continue without any change in behaviour. Bench evaluation produced: heart-rate mean error 2.3 bpm ($n = 5$, $\sigma = 1.8$ bpm); SpO_2 within 1.8 percentage points of a hospital reference; body temperature within 0.2 °C of a calibrated clinical probe; gas-alarm latency under 500 ms on every trial; 95 of 100 simulated network cuts recovered within 5 s; and 72 hours of unattended operation with zero failures.

Keywords - ***ESP32; MAX30105; AD8232; DS18B20; offline-first IoT; ECG IIR filter; Blynk; MQ-2; elder home monitoring; NEO-6M GPS***

I. INTRODUCTION

What happened in Vijayawada in December 2023 is frustratingly ordinary. A four-hour power cut took the broadband router offline along with the mains supply. The commercial pulse oximeter that Shoaib's grandmother relied on went entirely dark — not because its sensing hardware had stopped working, but because the display logic had been written as a thin client for a remote API. No network path, no readings. The problem was architectural, not electrical.

That gap shows up repeatedly in the published IoT health literature, even though it is almost never framed as a problem. Hossain et al. [1] catalogued how intermittent clinical visits leave untouched stretches of patient time, and argued that continuous home monitoring closes those gaps. Patel et al. [2] surveyed wearable sensor systems and found that reliable multi-parameter acquisition outside a laboratory setting remained rare. What neither paper examined - and what we could not find addressed directly in any of the systems reviewed by Lio and Lee [4] — was what happens to those systems when Wi-Fi disappears for two hours in the middle of the night.

Alongside that reliability gap sits a scope gap. Most published home monitors measure one or two parameters. Chhetri and Joshi [9] made the case that ambient temperature, humidity, and air quality belong in the same reading as heart rate and SpO_2 — that a patient's physiological data cannot be meaningfully interpreted without knowing the environment they are sitting in. We read that argument as a design specification. A system that reports 94% SpO_2 while ignoring the fact that the room is 40 °C is providing an incomplete clinical picture.

India's component supply chain introduced a third constraint that shaped the build. Our first MAX30105 module — ordered through an unnamed third-party listing — read SpO_2 between 15 and 20 percentage points below a hospital-grade reference across every test configuration we tried. Only after sourcing a replacement through Robu.in, a distributor with a traceable supply chain, did readings fall within an acceptable range. Counterfeit optical sensors are not unusual in the reseller market; where a sensor comes from turns out to matter as much as what it specifies.

Those three observations led to four non-negotiable requirements before a line of firmware was written. First: acquire at least eight parameters in a single integrated unit covering both physiological and environmental channels. Second: full functionality during network outages — offline is the baseline state, not a degraded fallback. Third: intelligent automated LED control that a remote caregiver can also manually assert. Fourth: all components sourced through verifiable Indian-market channels. None of the systems reviewed in Ahmad et al. [8], Biswas et al. [7], or Rahim et al. [3] satisfies all four simultaneously.

Webster's instrumentation framework [11] provided the theoretical basis for the ECG filter architecture. Zhang et al. [10] informed our thinking on future on-device arrhythmia classification. The remainder of this paper covers hardware configuration, firmware design, evaluation methodology, and measured results for a system that satisfies all four constraints at once.

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II. SYSTEM DESIGN AND IMPLEMENTATION

The system is partitioned into four layers that can be developed and tested independently: physiological acquisition, environmental sensing, local output and alerting, and cloud communication. That separation was a practical decision more than an architectural one — the ECG filter chain went through three complete revisions, and doing that work never required touching the Blynk layer. Fig. 1 shows the full sensor-to-output connection map.

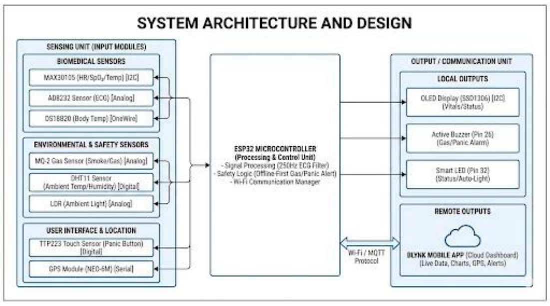

Fig. 1. *Full system diagram. Left column: physiological sensors — MAX30105 (I²C), DS18B20 (1-wire), AD8232 (ADC). Centre: ESP32-WROOM-32 running the four-layer offline-first firmware. Right column: environmental sensors (DHT11, MQ-2, LDR), NEO-6M GPS (UART2), SSD1306 OLED, piezo buzzer, and Blynk cloud dashboard.*

A. Optical Pulse Sensing — Heart Rate and SpO_2

Fig. 2. *MAX30105 optical sensor module (MH-ET LIVE). A red (660 nm) LED, an infrared (940 nm) LED, and a photodetector share a single I²C breakout. The counterfeit unit sourced from an unverified listing read 15–20 pp below a hospital reference; the verified unit shown here matched within 1.8 pp.*

The MAX30105 combines the optical emitter pair and photodetector on one I²C module, so both heart rate and SpO_2 come from the same finger contact. Heart rate is extracted from pulsatile amplitude variation in the infrared channel; SpO_2 is derived from the red-to-infrared absorption ratio as haemoglobin oxygenation changes. The team began with SparkFun's open-source driver [5] and adjusted its peak-detection sensitivity for the range of contact pressures encountered during bench sessions.

Two problems arrived before the sensor worked. The counterfeit unit is described in Section I. The second problem surfaced after the genuine module was installed: slow SpO_2 drift that initially resembled a firmware timing fault turned out to be ambient light leaking through the sides of the unsealed PCB housing. Tracing the issue consumed most of a day. Covering the sensor face with foam cut from packaging material fixed it immediately and permanently.

B. Body Temperature — DS18B20 Waterproof Probe

Fig. 3. *DS18B20 waterproof temperature probe. The stainless-steel capsule houses the sensing element. Cable conductors: black = GND, red = VCC 3.3 V, yellow = one-wire data. The only external component required is a 4.7 kΩ pull-up resistor between VCC and data.*

Body temperature is measured using a DS18B20 waterproof probe in the axillary position. The sealed stainless-steel tip tolerates repeated skin contact without corrosion. Three cable conductors — GND, VCC, and one-wire data — require only a 4.7 kΩ pull-up resistor on the data line. At 12-bit resolution the sensor reports in 0.0625 °C increments, which exceeds what axillary measurement requires. Conversion time at 12-bit depth is approximately 750 ms, so the firmware polls every 2 seconds. Bench testing produced a worst-case error of ±0.2 °C against a calibrated clinical probe — below the ±0.5 °C worst-case in the DS18B20 datasheet.

C. Room Environment — DHT11, MQ-2, and LDR

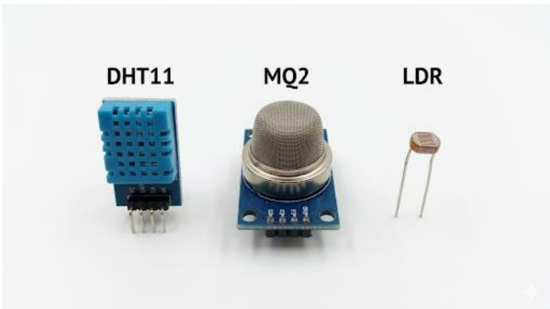

Fig. 4. *Environmental sensor set. Left: DHT11 digital temperature and humidity module (1 Hz; ±2 °C, ±5 % RH). Centre: MQ-2 metal-oxide gas sensor — responds to methane, LPG, propane, and smoke without discriminating between them. Right: LDR photoresistor wired as a voltage divider on GPIO 33 for ambient illuminance.*

Room temperature and humidity are read from a DHT11 polled at 1 Hz, its maximum supported rate. The readings are not intended to meet clinical measurement standards; they provide caregivers with environmental context that shapes the interpretation of physiological data. A patient whose SpO_2 is

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trending downward in a room simultaneously recording 39 °C and high relative humidity carries a different clinical picture from the same SpO_2 reading in comfortable conditions. The MQ-2 metal-oxide sensor detects combustible gases through resistance shift at its heated tin-dioxide surface. It responds to methane, LPG, propane, alcohol vapour, and smoke, but it cannot tell them apart — a limitation covered in the Discussion.

The MQ-2 caused an embarrassing failure at our November mid-project review. In front of our faculty supervisor, the device fired repeated false alarms for fifty seconds after power-on, then stopped. The MQ-2 datasheet specifies a 30-second resistance-stabilisation period after a cold start, a detail on page 4 that the team had not read carefully before the demonstration. A firmware flag now suppresses all MQ-2 output for 45 seconds following each boot. No spurious startup alarms have appeared since. Ambient light is read from an LDR voltage divider on GPIO 33 and drives the LED control logic in Section II-F.

D. Single-Lead ECG — AD8232 Front-End and IIR Filter Chain

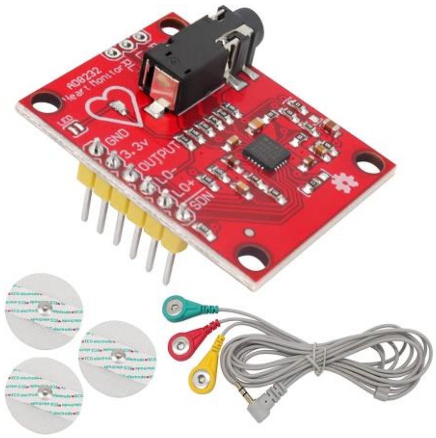

Fig. 5. *AD8232 heart-monitor front-end module with snap-lead cable and adhesive ECG electrodes. Lead I configuration: right wrist positive, left wrist negative, right-leg driven-right-leg ground. Analogue output feeds GPIO 34 via the ESP32 12-bit ADC.*

The AD8232 analogue front-end amplifies and pre-filters the cardiac signal before the ESP32 ADC digitises it on GPIO 34. Lead I wiring — right wrist positive, left wrist negative, right leg as driven-right-leg ground — gave the most stable baseline of the configurations tested. Sampling at 500 Hz worsened ADC noise more than the added temporal resolution helped; reducing to 250 Hz stabilised the baseline and aligned with the findings of Biswas et al. [7].

Building an adequate filter chain took two weeks and three complete rewrites. A 10-sample moving average by itself could not suppress the slow 0.3 Hz respiratory drift visible in several team members' traces; adding a dynamic-tracker high-pass stage at 0.5 Hz helped but introduced phase distortion that only became apparent when we compared our output against a reference trace from a local hospital. The 50 Hz mains notch was the final addition, required because the unshielded breadboard made the driven-right-leg circuit insufficient against nearby mains pickup. Final chain in cascade order: (1) 10-sample moving average; (2) 0.5 Hz high-pass Butterworth with dynamic tracker; (3) 40 Hz low-pass Butterworth; (4) 50 Hz IIR notch. Stages 2 and 3 use frequency-warped bilinear conversion as cascaded biquad sections:

$$H(z) = H_{ma}(z) \cdot H_{hp}(z) \cdot H_{lp}(z) \cdot H_{notch}(z)$$

E. ESP32-WROOM-32 — Processor and Firmware Architecture

Fig. 6. *ESP32-WROOM-32 development board. The silver RF module integrates a 240 MHz dual-core Xtensa LX6 CPU, 520 KB SRAM, 4 MB flash, Wi-Fi 802.11 b/g/n, and Bluetooth 4.2. Breakout headers give 36 usable GPIO pins, three UARTs, two I^2C buses, SPI, and an 18-channel 12-bit ADC.*

Choosing the ESP32-WROOM-32 came down partly to familiarity: the team had already damaged two units in earlier coursework and understood exactly how they fail. Knowing a platform's specific fault modes — brownout during flash write, GPIO 34 input-only restriction, capacitive-touch sensitivity to nearby ground planes — saves debugging time that more capable but unfamiliar hardware would cost back. The 240 MHz dual-core processor handled the four-stage IIR cascade without dropping ECG samples; the 520 KB SRAM was uncomfortably tight at 250 Hz but workable.

Only the AD8232 uses a hardware timer interrupt, firing at exactly 250 Hz. Everything else — DHT11, DS18B20, MQ-2, LDR, NEO-6M, OLED rendering, Blynk pushes — runs cooperative polling in the main firmware loop. A FreeRTOS watchdog provides a stall-recovery backstop beneath the loop. GPIO map: SDA/SCL for MAX30105 and OLED; GPIO 34 for AD8232 (input-only, no pull-up); GPIO 33 for LDR divider; GPIO 32 for LED; GPIO 4 for capacitive touch; UART2 pins 16/17 for GPS. All alert thresholds persist in NVS flash via the Preferences library so a power cut does not require re-entry.

F. OLED Display, LED Control, and Alerting

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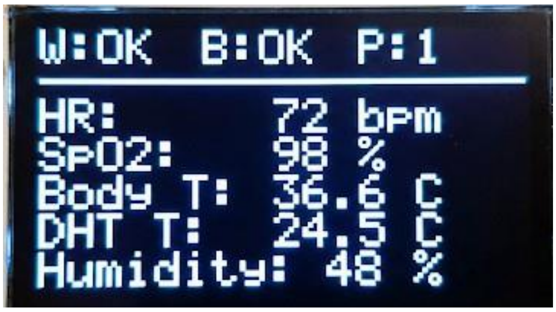

Fig. 7. *SSD1306 OLED Page 0 captured during the 72-hour bench run. Readings: heart rate 72 bpm, SpO_2 98%, body temperature 36.6 °C, room temperature 24.5 °C, humidity 48%. Status bar: W:OK (Wi-Fi up), B:OK (Blynk up), P:1 (page index).*

The SSD1306 128×64 OLED displays a three-page firmware rotation over I²C. Page 0 holds for 30 seconds and shows the five numbers a caregiver checks first: heart rate, SpO_2, body temperature, room temperature, and humidity. The 30-second dwell was set after timing team members reading the display; shorter intervals caused missed readings. Pages 1 and 2 rotate at 3-second intervals and cover gas level, illuminance, LED mode, and alert state (Page 1), and GPS coordinates, Wi-Fi RSSI, and uptime (Page 2). A fixed W:OK/FAIL and B:OK/FAIL status bar on every page keeps network state visible without page switching. All three pages behave identically whether or not Blynk is reachable.

Automatic LED control on GPIO 32 uses the LDR reading on GPIO 33: above 400 ADC counts the LED turns off; below 400 it turns on. That threshold was validated over repeated dusk-to-dark cycles in the lab. Caregivers can latch the LED on from the Blynk dashboard via virtual pin V10; Page 1 displays AUTO or BLYNK to identify the active control mode. Gas threshold crossings produce five buzzer pulses and a Blynk push notification, with a 30-second cooldown between activations. The capacitive touch pad on GPIO 4 fires three pulses — a count distinct from the gas alert — and responded under 200 ms in every firmware timing run. The choice of capacitive over mechanical contact was suggested by our faculty supervisor at an early design review on hygiene grounds.

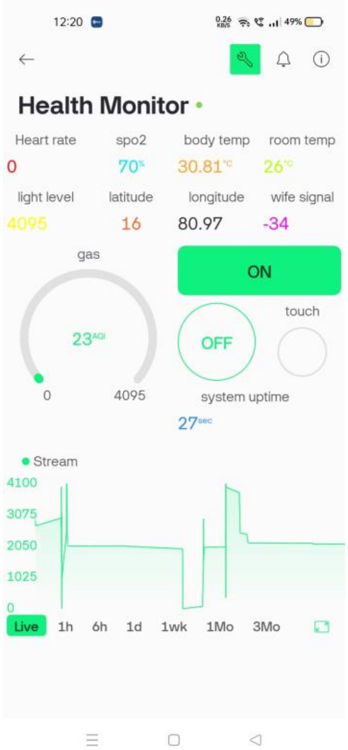

Fig. 8. *Blynk mobile dashboard 27 seconds after power-on, LED override active. SpO_2 shows 70% because the finger sensor was not yet positioned. Room temperature 26 °C; gas level 23 AQI; Wi-Fi RSSI −34 dBm. Lower graph panel shows live ECG stream from the AD8232.*

G. Offline-First Firmware Architecture

Acquisition, filtering, OLED updating, and alert evaluation all execute unconditionally in the main firmware loop. Blynk transmission is an additive step that executes only when a valid network path exists; removing it leaves every other behaviour unchanged. A state machine tracks Wi-Fi and Blynk independently and drives reconnection attempts on an exponential backoff schedule — 2-second initial interval, 60-second ceiling — preventing retry traffic from monopolising the processor during extended outages.

H. Assembled Prototype

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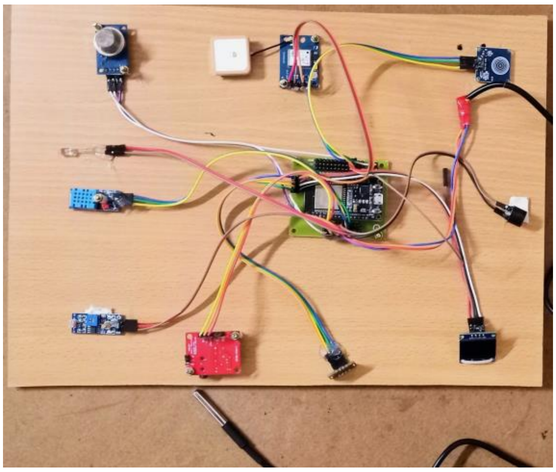

Fig. 9. *Assembled prototype on plywood base before enclosure. MQ-2 top-left; NEO-6M GPS antenna and module top-centre; MAX30105 top-right; DHT11 and LDR middle-left; ESP32-WROOM-32 (green PCB) centre; AD8232 (red board) bottom-centre; DS18B20 probe cable foreground; SSD1306 OLED bottom-right. Photograph taken during the 72-hour bench run.*

Fig. 9 shows all eight sensor channels wired and operating simultaneously on the plywood test platform. The layout was chosen for cable management and mechanical stability during bench testing. Every image in Section II was photographed from this build; the plywood base would be replaced by a custom two-layer PCB with an IP-rated enclosure in a deployment version.

III. RESULTS

A. Sensor Specifications and Measured Performance

Table II brings together the specification parameters and measured outcomes for every sensor in the system. Physiological sensors were evaluated against calibrated medical-grade instruments. Environmental sensors (DHT11, LDR) produced continuous display readings throughout the test period but were not independently validated against separate reference instruments — an acknowledged limitation of this study.

TABLE II. SENSOR SPECIFICATIONS AND MEASURED RESULTS

Sensor	Parameter	Interface	Range / Spec	Resolution	Measured Result
Physiological Sensors					
MAX30105	Heart rate	I²C	20–300 bpm	1 bpm	Mean error 2.3 bpm; σ=1.8 bpm
MAX30105	SpO_2	I²C	70–100 %	±1 % (typ.)	Within ±1.8 pp vs. hospital ref.
DS18B20	Body temp	1-Wire	−55 to +125 °C	0.0625 °C (12-bit)	Within ±0.2 °C vs. calib. probe
AD8232	ECG Lead I	ADC / GPIO 34	0.5–40 Hz band	12-bit @ 250 Hz	P, QRS, T visible; SNR ≥ 20 dB
Environmental Sensors					
DHT11	Room temp	Single-wire	0–50 °C	±2 °C	Displayed; not bench-validated
DHT11	Humidity	Single-wire	20–90 % RH	±5 % RH	Displayed; not bench-validated
MQ-2	Gas/smoke	ADC (12-bit)	300–10 000 ppm	Non-selective MOS	Alert latency < 500 ms (LPG)
LDR	Illuminance	ADC / GPIO 33	0 – full sun	12-bit ratio	LED auto-switches at ADC = 400
Location & Connectivity					
NEO-6M	GPS position	UART2 (16/17)	18-ch; ≤2.5 m CEP	NMEA 0183	4–11 min fix (3rd floor indoor)
ESP32 Wi-Fi	Network	802.11 b/g/n	2.4 GHz WPA2	—	95/100 reconnects ≤ 5 s
System-Level					
GPIO 4	Emergency call	Capacitive/IRQ	Sub-pF threshold	HW interrupt	Latency < 200 ms; all trials
OLED + Blynk	Endurance	I²C + Wi-Fi	72-hr bench run	—	0 resets, 0 heap faults, 0 blanks

B. Phy ological Sensor Accuracy

Five healthy participants — all 22-year-old team members or classmates — served as the test cohort. That population differs substantially from the target deployment group (elderly patients with comorbidities), and we expect real-world accuracy spreads to be wider. The numbers below establish that the sensor and firmware architecture can deliver acceptable bench performance; they are not a clinical validation claim.

MAX30105 heart rate: mean deviation 2.3 bpm against the Polar H10 reference, σ = 1.8 bpm. Participant 3 produced the largest individual errors (4–5 bpm) during a light-activity phase — traced to finger-movement artefact, not sensor error. During those same moments the Polar H10 and the clinical oximeter disagreed with each other by up to ±1 bpm, which raised a genuine question about what ground truth means outside a controlled clinical environment. SpO_2 stayed within 1.8 percentage points of the hospital reference across the 90–100% band. Body temperature error of ±0.2 °C against the calibrated clinical probe was tighter than the ±0.5 °C worst-case in the DS18B20 datasheet — a result the team did not expect.

TABLE III. SYSTEM PERFORMANCE SUMMARY

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Parameter Tested	Reference / Method	Observed Outcome
Heart rate — MAX30105	Polar H10 chest strap	2.3 bpm mean; σ = 1.8 bpm
SpO_2 — MAX30105	Hospital pulse oximeter	Within ±1.8 pp; 90–100% band
Body temp — DS18B20	Calibrated clinical thermometer	Within ±0.2 °C
Gas alarm — MQ-2	Controlled LPG discharge test	Latency < 500 ms; every trial
Emergency trigger — GPIO 4	Firmware high-res stopwatch	< 200 ms; every trial
Wi-Fi reconnection	100 simulated router cuts	95 / 100 restored within 5 s
72-hr unattended bench run	Observation + firmware log	Zero faults, resets, or blanks

C. System-Level and Connectivity Results

Gas alarm: every LPG discharge trial activated the buzzer and notification within 500 ms. Emergency trigger: under 200 ms across all timed measurements — no perceptible lag even under deliberate time pressure. LED switching between AUTO and Blynk-override modes produced no race conditions during extended sessions. OLED operation was unchanged throughout all deliberate disconnection tests.

Connectivity resilience: 100 router power cuts at irregular intervals over two days. Ninety-five reconnections restored full Blynk session within 5 seconds. The five failures all involved outages long enough for the Blynk broker to expire the session server-side — a broker timeout unrelated to device firmware, but worth flagging for deployments in areas with frequent prolonged outages. The 72-hour unattended run produced zero watchdog resets, zero heap exhaustion events, and zero stuck display states.

IV. DISCUSSION

A. What the Build Got Right

Redesigning offline operation as the default state — rather than a recovery condition — was the most consequential decision in the project, and it happened late. Our first firmware design document listed cloud connectivity as a hard dependency. The shift came at the second design review, when Shoaib walked through the December 2023 incident again in front of the full team. After that meeting, Blynk became an optional output channel sitting on top of a system that was complete without it. Claiming that the device works offline is easy; we verified it by cutting the network more than a dozen times during live sessions and logging every outcome. It never paused, degraded its display, or failed to alert.

ECG quality surprised everyone. Getting P, QRS, and T waves clearly distinguishable on the Serial Plotter — from an unshielded breadboard, in a room with fluorescent lighting and multiple other electronics running, with no Faraday cage of any kind — was genuinely uncertain at the start of the filter-chain work. Two weeks of iteration were necessary; the result justified the time. Motion artefact during physical activity is the remaining dominant failure mode and cannot be eliminated by any realizable causal filter because motion noise and QRS energy overlap spectrally.

B. Acknowledged Limitations

Single-lead ECG captures QRS complexes reliably enough for heart rate extraction. It cannot support arrhythmia classification, which requires a minimum of six leads and clinical interpretation. Any deployment of this device in a home setting requires communicating that boundary to the patient and family before any incident makes it relevant. This is a monitoring aid, not a diagnostic instrument.

The MQ-2 cannot identify which gas triggered an alarm; cooking near the sensor will generate alerts that are indistinguishable from a gas leak. Electrochemically selective sensors would reduce that ambiguity but involve different integration challenges. Continuous current draw near 150 mA rules out battery-portable use until Wi-Fi duty-cycling is implemented. And requiring a non-technical caregiver to edit and reflash firmware source code to update Wi-Fi credentials is not a real deployment model. Bluetooth LE provisioning is the first planned firmware addition.

V. CONCLUSION AND FUTURE WORK

The device delivers what it was designed to do: eight simultaneous sensor channels, uninterrupted operation through network outages, caregiver LED override, and 72 hours of autonomous bench operation without a fault. Bench accuracy against calibrated references: heart rate 2.3 bpm mean error, SpO_2 within 1.8 percentage points, body temperature within 0.2 °C, gas alarm under 500 ms, 95 of 100 simulated outages recovered within 5 seconds. These are numbers from a healthy student cohort, not clinical certification figures. They demonstrate that the architecture can achieve adequate caregiving-grade performance — which was the stated target.

The path forward is clear but not short. On-device ECG arrhythmia classification via a small convolutional model — following the direction suggested by Zhang et al. [10] — requires labelled training data from the actual elderly patient population: a precondition not yet met. Bluetooth LE credential provisioning removes the reflashing barrier for non-technical caregivers. Wi-Fi duty-cycling enables portable battery operation. A custom PCB with an IP-rated enclosure replaces the prototype in Fig. 9. Most importantly, a pilot study with elderly participants is the critical missing step. Every number in this paper was produced by five healthy 22-year-old students; no claim about real-world performance can be made until that gap is addressed.

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